SENTIMENT ANALYSIS OF McDONALD'S CUSTOMER REVIEWS

Final Report

DS4110 Aritificial Intelligence

Submitted by

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Index No: 22CDS0412

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August, 2026

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Chapter 1: Introduction

This project builds and evaluates a sentiment-classification system for McDonald's Google Maps customer reviews, comparing supervised machine-learning models (Logistic Regression, Linear SVM, Naive Bayes) against a lexicon-based VADER baseline. Objectives were to construct a labelled sentiment dataset validated through human annotation, train and compare classifiers using a class-imbalance-robust metric, and report the champion model's performance; scope covers data acquisition through model evaluation, excluding production deployment and multi-domain generalisation.

Chapter 2: Background and Problem Domain

Sentiment analysis classifies text polarity (Negative/Neutral/Positive); customer reviews are short, informal, and often rating-anchored, making rating-derived labels a natural but imperfect proxy for true sentiment — a 3-star review, for instance, may carry strongly negative wording that a star count alone would miss. Lexicon methods such as VADER require no training data but struggle with domain-specific and mixed-sentiment language, while supervised models require labelled data but generally achieve higher accuracy once trained; both approaches are therefore compared in this project against a human-verified ground truth.

Chapter 3: Methodology

3.1 Dataset

33,396 Google Maps reviews of McDonald's outlets were scraped and cleaned to 22,366 usable free-text reviews (5,167 auto-generated placeholder "reviews" were identified and excluded to prevent label leakage). Each record contains the review text, a 1–5 star rating, and outlet metadata.

3.2 Labeling Pipeline

Ground-truth labels were originally rating-derived (1–2→Negative, 3→Neutral, 4–5→Positive). To validate this proxy, three annotators independently labelled a stratified 1,000-review pilot from text alone (blind to rating). Inter-annotator agreement was substantial (Fleiss' $\kappa = 0.629$; 66.4% full agreement). The resolved human labels were compared against the rating-derived proxy: accuracy 0.789, Cohen's κ 0.654, macro-F1 0.694 — below the pre-set 0.80 acceptance bar, so the rating proxy alone was not adopted.

A hybrid pseudo-labeling approach was used instead. The 1,000 human labels were split 700/150/150 (train/validation/gold-test). A TF-IDF (1–2-gram) + Logistic Regression model (C=2.0, class_weight='balanced') was tuned on the validation split and evaluated once on the untouched gold-test set: accuracy 0.707, macro-F1 0.599. This model was refit on all 850 human train+validation rows and used to pseudo-label the remaining 21,371 reviews (9,522 of these, 44.5%, fell below a 0.60 confidence threshold and were flagged needs_human_review).

Methodological caveat: the final training/test pool is therefore 995 human-verified labels (4.4%) plus 21,371 model-generated pseudo-labels (95.6%). Downstream model accuracy in Section 3.2 is measured mostly against pseudo-labels, not independent human ground truth, and should be read alongside the smaller gold-test figures above, which are the only fully human-verified evaluation in this project.

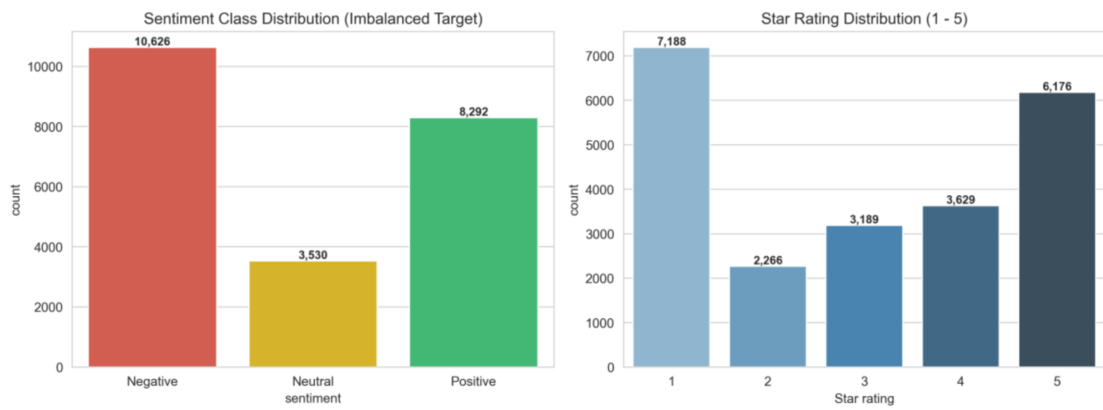

Figure A: Sentiment class distribution after rating-derived labelling (22,366 reviews).

3.3 Preprocessing Steps

Step	Detail
Noise removal	Strip URLs, emojis, HTML artefacts, non-alphabetic tokens
Normalisation	Lower-casing; expand contractions
Auto-tag filtering	Remove 5,167 platform-generated placeholder reviews
Tokenisation	Negation-preserving word tokeniser (keeps "not_good" as one unit)
Stopword handling	Domain stopword list; sentiment-bearing negators retained
Vectorisation	TF-IDF, 1–3-grams, 20,000 features, min_df=3, sublinear TF

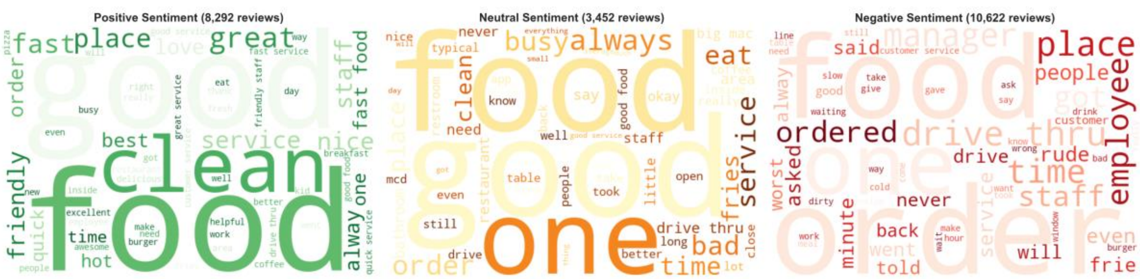

Figure B: Word clouds by sentiment class after preprocessing.

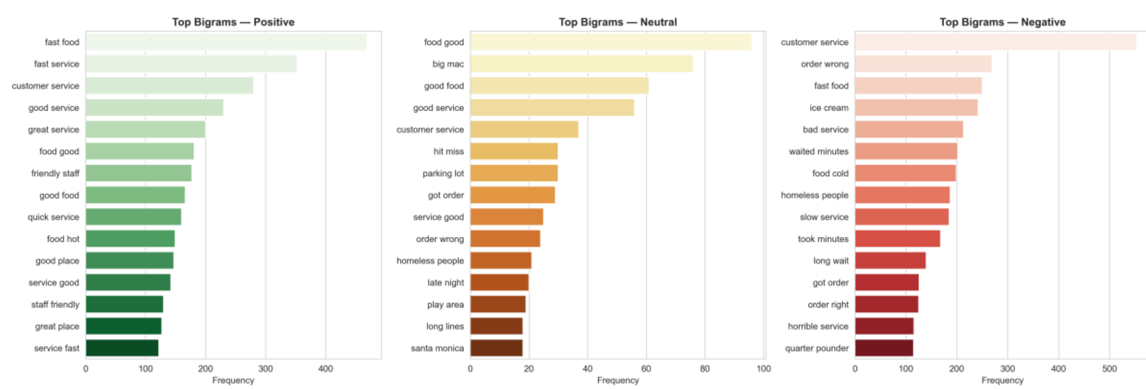

Figure C: Most frequent bigrams by sentiment class.

3.4 Feature Engineering & Models

Review text was cleaned (lower-casing, URL/punctuation handling, negation-preserving tokenisation) and vectorised with TF-IDF (unigrams–trigrams, 20,000 features, min_df=3, sublinear TF). Three supervised classifiers were trained with class_weight='balanced' to counteract Neutral-class under-representation: multinomial Logistic Regression, Linear SVM (LinearSVC), and Multinomial Naive Bayes. An unsupervised VADER lexicon baseline was run on an 8,000-review sample for comparison, requiring no training data.

3.4b Train / Test Split

Stratified 80/20 split (17,892 train / 4,474 test) on the full labelled pool, preserving class proportions; the same split was used to evaluate all four approaches for comparability.

3.5 Evaluation Metrics

Metric	Definition
Accuracy	Proportion of all test reviews correctly classified
Macro Precision	Unweighted mean of per-class precision
Macro Recall	Unweighted mean of per-class recall
Macro F1	Unweighted mean of per-class F1; primary model-selection metric under class imbalance

Chapter 4: Results, Insights and Discussion

4.1 Model Comparison

Champion model: Linear SVM, selected as the model with the highest macro-F1 (0.839), also achieving the highest accuracy (87.2%).

Model	Accuracy	Macro Prec.	Macro Rec.	Macro F1
Linear SVM (best)	0.8722	0.8370	0.8414	0.8391
Logistic Regression	0.8663	0.8279	0.8544	0.8369
Multinomial Naive Bayes	0.7827	0.8303	0.6318	0.6114
VADER (unsupervised)	0.6305	0.5843	0.5847	0.5699

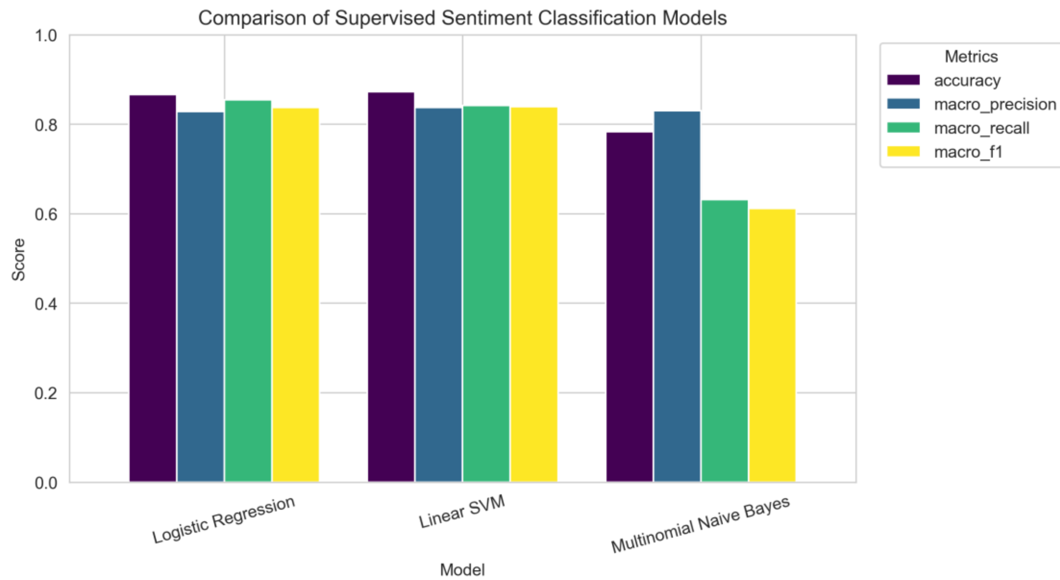

Figure 1: Accuracy, macro-precision, macro-recall and macro-F1 across all four approaches.

4.2 Per-Class Performance — Linear SVM

Class	Precision	Recall	F1-score	Support
Negative	0.90	0.90	0.90	2,125
Neutral	0.70	0.73	0.71	690
Positive	0.91	0.89	0.90	1,659

Naive Bayes reaches the highest macro-precision (0.830) but the lowest macro-recall (0.632) and macro-F1 (0.611): it almost never predicts "Neutral", defaulting to the majority classes — the reason raw accuracy alone is a misleading selection criterion here. Linear SVM and Logistic Regression track each other closely on every metric (within 0.005), with SVM ahead on all four; Neutral remains the hardest class for both, consistent with its smaller support (690 of 4,474 test reviews) and its semantic overlap with mild positive/negative language.

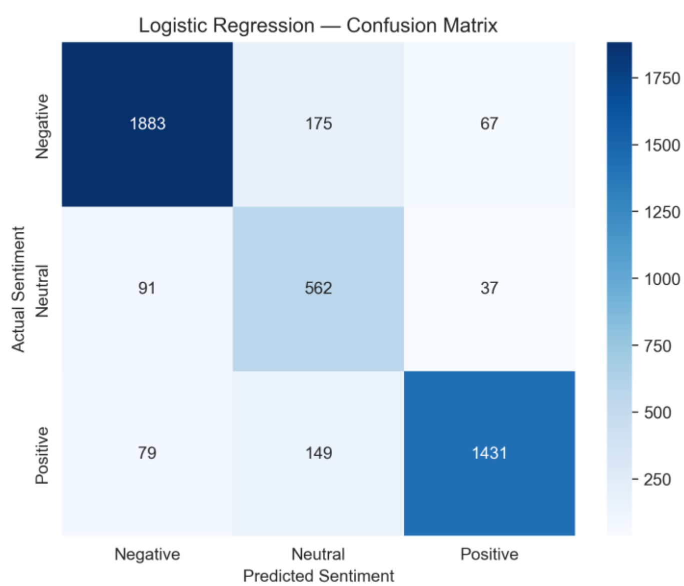

Figure 2: Confusion matrix, Logistic Regression (retained for coefficient interpretability alongside the SVM champion).

Actual \ Predicted	Negative	Neutral	Positive
Negative	1,893	175	67
Neutral	91	562	37
Positive	79	149	1,431

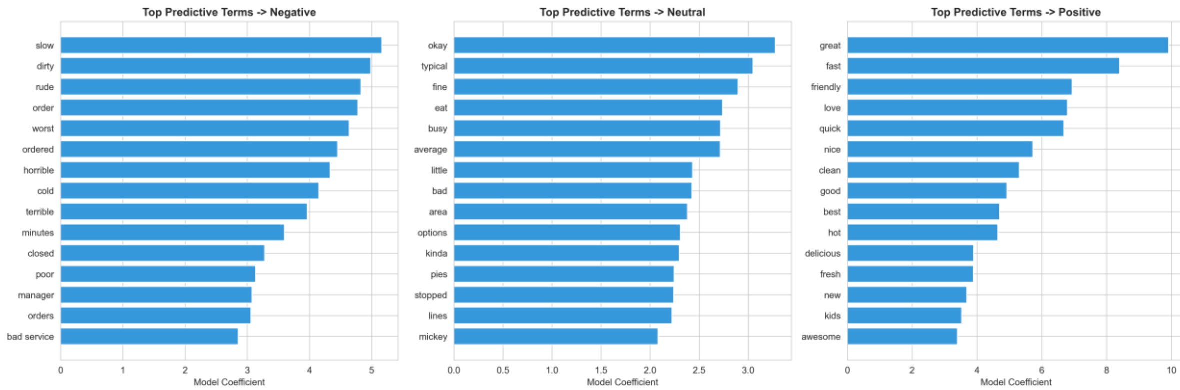

Figure 3: Top TF-IDF-coefficient words per sentiment class (Logistic Regression).

4.3 Human-Labeling Validation (Gold Test, n=150)

Metric	Value
Accuracy	0.7067
Macro F1	0.5986
Fleiss' κ (pilot IAA)	0.6287

This is the only fully human-verified evaluation in the project; it is markedly lower than the pseudo-label-based test accuracy in Section 2.1, consistent with the caveat in Section 1.2.

4.4 VADER Lexicon Baseline

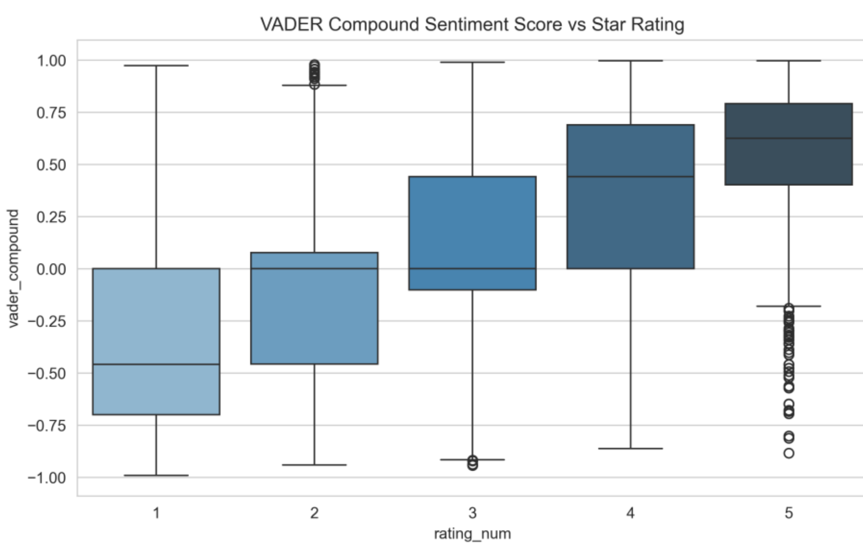

Figure 4: VADER compound score vs. star rating — monotonic trend supports the rating-derived labelling scheme.

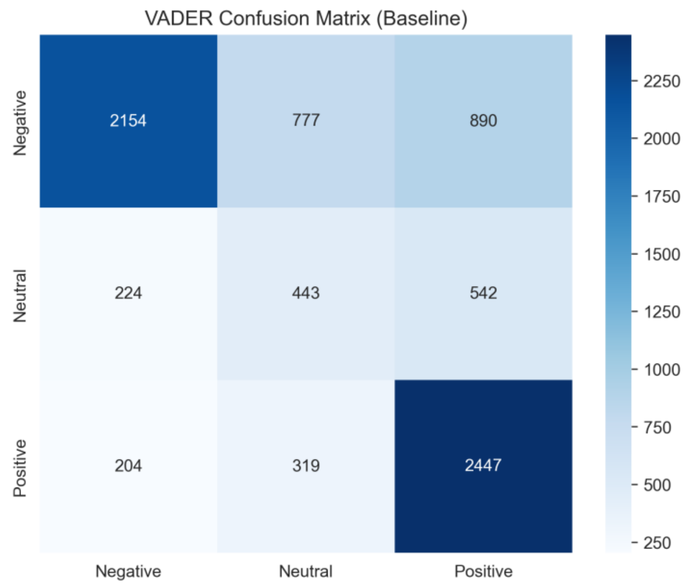

Figure 5: VADER confusion matrix (8,000-review sample).

VADER reached 63.1% accuracy / 0.570 macro-F1 with zero training data, versus the champion Linear SVM's 87.2% / 0.839 — quantifying the value of labelled training data, though noting the SVM figure includes the pseudo-label caveat above.

4.5 Full Per-Class Metrics — All Supervised Models

Model	Class	Precision	Recall	F1
Linear SVM	Negative	0.90	0.90	0.90
Linear SVM	Neutral	0.70	0.73	0.71
Linear SVM	Positive	0.91	0.89	0.90
Logistic Regression	Negative	0.90	0.89	0.90
Logistic Regression	Neutral	0.64	0.81	0.72
Logistic Regression	Positive	0.92	0.87	0.89
Multinomial NB	Negative	0.79	0.94	0.86
Multinomial NB	Neutral	0.85	0.03	0.06
Multinomial NB	Positive	0.78	0.87	0.82

Logistic Regression's higher Neutral recall (0.81 vs. SVM's 0.73) comes at the cost of lower Neutral precision (0.64 vs. 0.70) — it over-predicts Neutral. SVM's more balanced precision/recall trade-off across all three classes is the main reason its macro-F1 edges out Logistic Regression despite similar accuracy. Naive Bayes' near-zero Neutral recall (0.03) confirms it is unsuitable for this three-class problem despite its high overall accuracy.

Chapter 5: Limitations and Future Work

- Test-set circularity: ≈95% of the evaluation labels used in Section 2.1 are model-generated pseudo-labels rather than independent human judgments (Section 1.2); the Section 2.3 gold-test figures are the more conservative, fully human-verified estimate.

- Neutral-class difficulty: smallest class by support (15.4% of test data) and semantically closest to mild Positive/Negative language, driving most residual error for every model.
- Confidence-flagged pseudo-labels: 44.5% of pseudo-labelled rows fell below the 0.60 confidence threshold and were never reviewed by a human, so their true label accuracy is unverified.
- Single-domain scope: all reviews are for one restaurant chain; generalisation to other domains is untested.

5.1 Future Enhancements

- Extend human annotation beyond the 1,000-review pilot to reduce reliance on pseudo-labels.
- Add geographic/outlet-level sentiment breakdowns for operational use.
- Explore transformer-based embeddings (e.g., DistilBERT) as a higher-capacity alternative to TF-IDF.

5.2 Deployment Note

The champion model (Linear SVM) and its TF-IDF vectoriser were packaged for batch inference; Logistic Regression is retained alongside it for coefficient-level interpretability of predictive terms.

Chapter 6: Conclusion

This project delivered a validated sentiment-labelling pipeline and a comparative evaluation of three supervised classifiers against a lexicon baseline. Linear SVM was selected as the champion model on macro-F1 (0.839) and accuracy (87.2%), narrowly ahead of Logistic Regression, with both far ahead of Naive Bayes and VADER. The human-labeling pilot (gold-test accuracy 70.7%, macro-F1 0.599) is retained as the project's most trustworthy evaluation and should be read alongside the headline pseudo-label-based figures, per the limitation noted in Chapter 5.

References

- *NLTK VADER Sentiment Analysis Tool — Hutto, C.J. & Gilbert, E. (2014).*
- *Scikit-learn: Machine Learning in Python — Pedregosa et al. (2011), JMLR.*
- *TF-IDF and n-gram feature engineering — standard scikit-learn text feature extraction documentation.*